8.3.1 The rise of autonomous combat and surveillance systems

The evolution of UAVs from manually piloted platforms to highly autonomous systems marks a pivotal shift in contemporary military strategy. Modern combat environments require rapid situational awareness, persistent surveillance, and precision engagement, all of which drones are uniquely equipped to provide. Advances in machine learning, pattern recognition, onboard processing systems, and sensor fusion have enabled drones to perform operations such as target identification, route optimization, surveillance patrolling, and battlefield reconnaissance with significantly reduced human intervention. Many state militaries now employ drones not only as tools for collecting intelligence but also as independent or coordinated strike units capable of engaging targets in contested environments.

The integration of UAVs into network-centric warfare has further transformed command and control hierarchies. Drones are now embedded into multi-layered battle networks where they act as data nodes collecting, transmitting, and interpreting real-time battlefield information. This has given rise to concepts such as “loyal wingman” systems, in which autonomous drones accompany crewed fighter jets to perform support or offensive roles, effectively multiplying air combat capabilities while minimizing risk to human pilots. Likewise, drone swarming frameworks allow dozens or hundreds of small UAVs to operate cooperatively, overwhelming air defense systems with distributed intelligence and coordinated movement (Ryan Jr, et al. 2019).

Additionally, the rise of autonomous surveillance drones has revolutionized information dominance, an increasingly decisive factor in modern warfare. High-altitude long-endurance (HALE) UAVs can patrol border regions, maritime stretches, and conflict zones continuously for